

The Renewable Energy Supply Chain for Data Centers: Opportunities, Challenges, and Future Directions

Introduction

Data centers play a crucial role in the digital economy, serving as the backbone of modern information technology infrastructure. The evolution of data centers dates back to the 1940s when government organizations began to leverage powerful computing power to support critical defense and intelligence projects (Digital Realty, n.d.). Over the decades, data centers have transformed significantly, becoming more efficient (smaller size, lower cost, more powerful, and less energy use) and scalable. They are commonly built to be the system running the entire enterprise IT infrastructure.

In the past 5 years, rapid advancements in blockchain, cloud computing, and artificial intelligence have driven a substantial increase in worldwide energy consumption. In the United States, data centers accounted for approximately 3% of electricity usage in 2022, with projections suggesting this could rise to 8% by 2030 (Goldman Sachs, n.d.). This surge in energy demand poses significant sustainability challenges, highlighting the urgent need for renewable energy solutions within the sector.

Compared to coal-fired power plants, electricity from renewable energy sources emits between 90 and 99% less greenhouse gases (GHGs) and produces 70 to 90% less pollutants (Times of India, 2024). Additionally, renewable energy can supply two-thirds of the total global energy demand, and contribute to the bulk of the greenhouse gas emissions reduction that is needed between now and 2050 for limiting average global surface temperature increase below 2 °C (Wong, Goh, & Chong, 2019). In this sense, transitioning to renewable energy is not just a choice but a necessity to ensure a sustainable future for data centers and the broader

environment. As the world grapples with the implications of climate change, embracing renewable energy is critical for mitigating its impact and aligning with global sustainability goals.

This paper aims to explore the renewable energy supply chain for data centers, focusing on the opportunities and challenges associated with this transition. By analyzing the current state of energy consumption in data centers and the significance of renewable energy, this research will provide insights into the strategies that can be employed to optimize the renewable energy supply chain. The significance of this research lies in its potential to inform stakeholders in the technology and energy sectors on the topic of integrating sustainable practices into data center operations. Ultimately, this paper will highlight future directions for enhancing the renewable energy supply chain, fostering innovation, and promoting sustainability in data center management.

Overview of the Renewable Energy Supply Chain

Renewable energy is energy derived from natural sources that are replenished at a higher rate than they are consumed (United Nations, n.d.). On top of that, the renewable energy supply chain is mainly consisting of five phases namely procurement, generation, transmission, distribution, and demand. These phases cover all processes along the supply chain of renewable energy, from raw materials (input) to the final product (output) (Scholarly Community Encyclopedia, n.d.). Each phase of the supply chain plays a vital role in ensuring the efficient delivery of renewable energy.

Various types of renewable energy sources are particularly relevant for powering data centers. Solar energy, for instance, harnesses sunlight through photovoltaic panels to generate

electricity. Wind energy captures the kinetic energy of wind through turbines to produce power. Hydropower, which utilizes the energy from flowing or falling water, contributes significantly to the renewable energy mix (National Grid, n.d.). Also, geothermal power plants extract fluids from underground reservoirs to produce steam or hot liquid, which drives turbines to generate electricity before being reinjected into the reservoir. (U.S. Department of Energy, n.d.). These sources not only provide clean energy but also help data centers mitigate their carbon footprint, reducing reliance on fossil fuels and enhancing planet Earth's development sustainability.

Key players in the renewable energy supply chain include suppliers, manufacturers, and distributors. Suppliers provide the necessary raw materials for energy systems like silicon for solar panels, metals for wind turbine blades, or specialized components needed for renewable energy systems, while manufacturers assemble the renewable energy components into finished products such as solar panels and wind turbines and produce the energy needed to use. Distributors play a critical role in the logistics and delivery of energy, ensuring that renewable resources reach consumers efficiently (Amigo Energy, n.d.). Prominent companies in this space include Schneider Electric, recognized for its role in facilitating power purchase agreements(PPA) and providing managed services (Schneider Electric, n.d.). As the demand for renewable energy grows, collaboration among these stakeholders becomes increasingly important for optimizing the supply chain and achieving sustainability goals.

Current Trends in Renewable Energy Adoption for Data Centers

By 2024, Microsoft, Google, and Amazon account for 60 percent of hyperscale capacity (Yadav, 2024). As a result, they are also leading the integration of renewable energy into their data center operations, driven by motivations such as cost savings, corporate social responsibility(CSR), and regulatory compliance. Google Cloud, Microsoft Azure, Amazon Web

Services (AWS) and Meta are all aiming to operate their data centers with 100% renewable energy between 2020 and 2025 (Cedara, 2023).

These major technology companies have become the biggest buyers of green energy by means of power-purchase agreements (PPAs). These are long-term contracts between a buyer (like a data-centre company) and a power producer (like a solar- or wind-farm operator) to buy electricity at a predetermined price, instead of bulk-buying from utilities (The Economist, 2024). In this way, PPA allows renewables projects to increase their level of revenue certainty. By providing financing, PPAs also enable the development of large-scale wind and/or solar projects, making the buyer's role essential for future renewable development (NextEra Energy Resources, 2023). This highlights that tech companies are not only top consumers of renewable energy but also pivotal contributors to its societal adaption development, directly influencing the acceleration of green energy infrastructure globally.

Moreover, many data center operators are leveraging Battery Energy Storage Systems (BESS) to mitigate the intermittency of wind and solar power and make better use of these renewables to reduce emissions while maintaining availability (Schneider Electric, 2024). The systems store electric power from renewable energy sources, thus providing backup power when needed and keeping data safe during events like power outages (Hanna et al., 2024). Compared with traditional diesel generators and their well-documented long-term negative impacts on the environment, BESS offers a cleaner and more sustainable alternative while ensuring operational resilience.

Challenges in the Renewable Energy Supply Chain for Data Centers

Despite the growing adoption of renewable energy in data centers, several challenges hinder the seamless integration of sustainable practices. One of the most significant is infrastructure. Renewable energy requires significant land use, with wind turbines needing even spacing across farms and solar plants occupying more space than traditional power plants, making them less efficient in land utilization. In contrast, when it comes to conversion efficiency, renewable technologies are only roughly half efficient compared with traditional energy devices (Inspire Clean Energy n.d.). This combination of high land requirements and lower conversion efficiency underscores the need for innovative solutions to optimize renewable energy systems for data center applications.

Further, as previously mentioned, the intermittent nature of renewable energy sources, such as solar and wind, necessitates advanced energy storage systems to ensure a steady power supply. However, according to Schneider Electric, as of May 2024, a Battery Energy Storage Systems (BESS) stores (typically) only one to two hours of energy in batteries. Its current use case mostly is just to help stabilize the grid, to provide additional backup power, not yet to equip the ability to replace the traditional energy backup device — diesel generators.

Finally, the rapid advancement of artificial intelligence (AI) technologies has intensified the energy demands of data centers, exposing critical challenges in the renewable energy supply chain. To begin, the AI model cannot understand the world just by examining how words and phrases appear together. To make inferences, it needs to be trained on massive amounts of data to observe as many combinations as possible (Numenta, 2023). All of these massive training and future user AI interaction processes need computing power support in data centers. As a result, it caused enormous energy use in data centers annually, which is estimated to be power consumption for approximately 21,602 U.S. homes for an entire year (RW Digital, 2023). As the

pace of efficiency gains in electricity use slows and the AI revolution gathers steam, Goldman Sachs Research estimates that data center power demand will grow 160% by 2030. This surge in energy consumption poses a significant barrier to incorporating net-zero energy practices in data center operations. Operators must accelerate the development of scalable renewable energy supply chains while simultaneously enhancing the efficiency of these renewable sources' energy conversion rates to meet the demands of the rapidly advancing AI era.

Future Directions and Innovations

As data centers continue to face mounting energy demands, the exploration of future directions and innovations in renewable energy is critical. One emerging avenue is the implementation of nuclear power. Nuclear power plants are producing maximum power more than 92% of the time during the year. That's about nearly 2 times more as natural gas and coal units, and almost 3 times or more reliable than wind and solar plants (U.S. Department of Energy, n.d.). While not classified as renewable, nuclear energy produces minimal carbon emissions and offers a reliable, high-output solution that can address the intermittency and low-efficiency challenges of renewable energy sources like solar and wind. Additionally, small Modular Reactors (SMRs), in particular, present an innovative option, providing scalable and safer nuclear power solutions that could be co-located with data centers to ensure a consistent energy supply (Schneider Electric, n.d.).

Regarding renewable energy sources, geothermal energy also holds immense potential for powering data centers, offering consistent and sustainable energy generation. It provides a stable, weather-independent energy supply for critical operations and requires minimal land use (TWI, n.d.). However, this promising energy source remains underused due to space limitations in urban areas, high upfront investment costs despite low operational expenses, and a lack of public

awareness about its widespread (CORDIS, n.d.). Future directions in geothermal technology include advancements in Enhanced Geothermal Systems (EGS), which enable energy extraction from less permeable rocks, significantly expanding the range of viable locations. Innovations such as hybrid systems combining geothermal with other renewable sources and direct geothermal cooling systems tailored for data centers are also gaining traction (Oil & Gas Portal, n.d.). These developments reduce carbon emissions, improve energy efficiency, and enhance operational sustainability. With continued investment and supportive policies, geothermal energy is poised to play a key role in the data center industry's shift towards carbon-free, reinforcing its positive outlook as a cornerstone of green computing.

Case Study: Renewable Energy Practices in Google Data Centers

Google, a leader in global technology and innovation, has long been at the forefront of adopting renewable energy practices to power its operations. Data centers, which are critical for Google's cloud services, consume vast amounts of energy, making sustainability initiatives in these facilities vital to achieving the company's environmental goals. As part of its commitment, Google aims to operate entirely on carbon-free energy by 2030, a goal that involves significant advancements in energy sourcing, management, and collaboration (Google Cloud, n.d.).

Google's ambition reflects its leadership in addressing climate change. To achieve this goal, Google has invested in numerous renewable energy projects worldwide. These projects directly power their data centers with renewable energy sources, such as wind and solar. For example, Google collaborated with a wind farm in Oklahoma to provide clean energy to its Pryor data center (Data Center Knowledge, n.d.). Similarly, investments in solar energy and wind projects in Denmark (Google Cloud, n.d.) contribute significantly to Google's efforts to decarbonize its operations. These localized investments allow Google to address energy

requirements sustainably while supporting renewable energy growth in the regions where it operates.

Beyond purchasing renewable energy, Google employs innovative technologies to enhance energy efficiency in its data centers. Leveraging machine learning algorithms and non-portable water resources, the company optimized its cooling systems, reducing electricity usage and enhancing operational efficiency. These data centers are among the most efficient globally, using 10% less energy than the industry average. In 2021, water cooling helped us reduce the energy-related carbon footprint of our data center portfolio by roughly 300,000 tons of CO₂ (Google, n.d.).

Google's approach also includes integrating battery storage technology to address the intermittency of renewable energy and potential power outage issues. At its St. Ghislain, Belgium, data center, batteries provide backup power and support grid balancing. This scalable solution contributes to Google's 24/7 carbon-free energy goal and advances sustainable data center operations (Google Cloud, n.d.).

Collaboration with utilities, governments, and other organizations plays an imperative role in accelerating Google data centers' transition to clean energy. Its partnerships often focus on expanding access to renewable energy in local grids, benefiting communities beyond its immediate operations. For instance, Google's clean energy projects contribute to job creation in renewable energy sectors, fostering economic growth while reducing environmental impact (Google, n.d.).

Personal Experience and Insights

My experiences and engagements have deeply influenced my interest in renewable energy and sustainability initiatives. My passion began during my co-op, where I managed the

company's EPA SmartWay project. Witnessing the dual impact of cost savings and environmental protection firsthand solidified my ambition to pursue a career focused on delivering green transformation solutions.

Returning to Northeastern, I joined the NEU Energy Systems Society to deepen my understanding of renewable energy and related fields. Through this organization, I explored the multifaceted benefits of renewable energy and engaged in discussions about the future of sustainable energy practices. I also participated in events and visits, such as volunteering at the NEU Energy conference and touring Northeastern's Snell Library rooftop photovoltaic system, which expanded my knowledge of the technical and practical applications of renewable energy systems and learned about the pressing issue of surging energy demand in data centers. These experiences inspired me to think deeply about how sustainable practices could be integrated into industries reliant on high energy consumption.

I hope that the intersection of technology, sustainability, and real-world impact can be the major theme of my future career. I will be happy to continue leading projects to reduce environmental footprints while optimizing operations in my future role. The knowledge I have gained from exploring renewable energy in data centers aligns with my future endeavors. By researching insights into renewable energy solutions like power purchase agreements, geothermal systems, and battery storage, I gained knowledge of popular industry practices and future innovations.

In my career, I aim to integrate green innovations into energy-intensive industries, particularly focusing on optimizing supply chains and implementing scalable renewable

solutions and this paper is conducive practice for me to build my knowledge base for achieving my goal.

Conclusion

In summary, the transition to renewable energy in data centers presents both significant opportunities and notable challenges. Key findings from the research indicate that renewable energy sources, such as solar, wind, hydropower, and geothermal, are essential for reducing the carbon footprint of data centers. As data centers increasingly integrate renewable energy technologies, they not only enhance their operational efficiency but also position themselves favorably in an environmentally conscious market.

However, several barriers remain, including infrastructure, battery limitations, and surging computing energy demand caused by the AI boom. Overcoming these challenges requires collaboration among stakeholders, including businesses, policymakers, and technology providers. Emerging technologies, coupled with supportive investments, can facilitate the adoption of sustainable practices and drive the industry toward a greener future.

As businesses and stakeholders in the industry, it is imperative to take action now. Embracing renewable energy not only contributes to environmental sustainability but also drives competitive advantage and long-term success.

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